16 There are only 3 red counters and 5 yellow counters in a bag.

Jude takes at random 3 counters from the bag.

Work out the probability that he takes exactly one red counter.

Exactly one red: RYY, YRY or YYR

P(RYY) = (3/8)(5/7)(4/6) = 60/336

P(YRY) = (5/8)(3/7)(4/6) = 60/336

P(YYR) = (5/8)(4/7)(3/6) = 60/336

Total = 180/336 = 15/28

1 mark
without-replacement second draw

1 mark
one correct branch product

1 mark
sum three valid orders

1 mark
15/28

15/28

(Total for Question 16 is 4 marks)

DO NOT WRITE IN THIS AREA
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